

Differential Equations Lesson 14

Question 1:

The differential equation $(x^2 - 4)y'' + 5xy' + x^2y = 0$, centered at the ordinary point $x = 1$, has the following power series solution:

$$y(x) = a_0 \left[1 + \frac{1}{6}(x-1)^2 + \frac{13}{54}(x-1)^3 + \frac{29}{108}(x-1)^4 + \frac{157}{540}(x-1)^5 + \dots \right] \\ + a_1 \left[(x-1) + \frac{5}{6}(x-1)^3 + \frac{53}{54}(x-1)^3 + \frac{59}{54}(x-1)^4 + \frac{217}{180}(x-1)^5 + \dots \right]$$

- a) The singular points of the differential equation $(x^2 - 4)y'' + 5xy' + x^2y = 0$ are _____?
- b) Radius of convergence for power series solution is _____?
- c) Interval of convergence for power series solution is _____?

Hint:

Write $(x^2 - 4)y'' + 5xy' + x^2y = 0$ as $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$.

If $P(x_0)$ is undefined or $Q(x_0)$ is undefined, then x_0 is a singular point.

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

Question 2:

The differential equation $(x^2 + 5x + 6)y'' + 2xy' + y = 0$, centered at the ordinary point $x_0 = 2$, has the following power series solution:

$$y(x) = a_0 \left[1 - \frac{1}{40}(x-2)^2 + \frac{13}{2400}(x-2)^3 - \frac{73}{9600}(x-2)^4 + \dots \right] \\ + a_1 \left[(x-2) - \frac{1}{10}(x-2)^3 - \frac{1}{300}(x-2)^3 + \frac{23}{6000}(x-2)^4 + \dots \right]$$

- a) The singular points of the differential equation $(x^2 + 5x + 6)y'' + 2xy' + y = 0$ are _____?
- b) Radius of convergence for power series solution is _____?
- c) Interval of convergence for power series solution is _____?

Hint:

Write $(x^2 + 5x + 6)y'' + 2xy' + y = 0$ as $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$.

If $P(x_0)$ is undefined or $Q(x_0)$ is undefined, then x_0 is a singular point.

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

Question 3:

The differential equation $(x^2 + 9)y'' + 2xy' + 4y = 0$, centered at the ordinary point $x_0 = 1$, has the following power series solution:

$$y(x) = a_0 \left[1 - \frac{1}{5}(x-1)^2 + \frac{1}{30}(x-1)^3 + \dots \right] + a_1 \left[(x-1) - \frac{1}{10}(x-1)^3 - \frac{1}{30}(x-1)^3 + \dots \right]$$

- a) The singular points of the differential equation $(x^2 + 9)y'' + 2xy' + 4y = 0$ are _____?
- b) Radius of convergence for power series solution is _____?
- c) Interval of convergence for power series solution is _____?

Hint:

Write $(x^2 + 9)y'' + 2xy' + 4y = 0$ as $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$.

If $P(x_0)$ is undefined or $Q(x_0)$ is undefined, then x_0 is a singular point.

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

Hint: The distance between two complex numbers, $z_1 = a_1 + b_1 \cdot i$ and $z_2 = a_2 + b_2 \cdot i$,

$$\text{is } d = \sqrt{(a_2 - a_1)^2 + (b_2 - b_1)^2}.$$

Question 4:

The differential equation $x^2 \cdot (x^2 - 4)y'' + 2xy' + 4y = 0$, centered at the ordinary point $x_0 = 1$, has the following power series solution:

$$y(x) = a_0 \left[1 - \frac{2}{3}(x-1)^2 - \frac{4}{27}(x-1)^3 + \dots \right] + a_1 \left[(x-1) + \frac{1}{3}(x-2)^3 + \frac{7}{27}(x-1)^3 + \dots \right]$$

- a) The singular points of the differential equation $x^2 \cdot (x^2 - 4)y'' + 2xy' + 4y = 0$ are _____?
- b) Radius of convergence for power series solution is _____?
- c) Interval of convergence for power series solution is _____?

Hint:

Write $x^2(x^2 - 4)y'' + 2xy' + 4y = 0$ as $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$.

If $P(x_0)$ is undefined or $Q(x_0)$ is undefined, then x_0 is a singular point.

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

Question 5:

Find the power series solution for the differential equation $x \cdot y'' + y' + y = 0$, centered at the ordinary point $x_0 = 4$.

Since we are looking for a power series solution centered at $x_0 = 4$, define $t = x - 4$.

Hence, $x = t + 4$.

$$\text{Note: } \frac{dx}{dt} = 1, \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{dy}{dt}, \text{ and } \frac{d^2y}{dx^2} = \frac{d}{dx} \cdot \left[\frac{dy}{dx} \right] = \frac{d \left[\frac{dy}{dx} \right] / dt}{dx/dt} = \frac{d \left[\frac{dy}{dt} \right] / dt}{1} = \frac{d^2y}{dt^2}.$$

$$\text{Write } x \cdot y'' + y' + y = 0 \text{ in terms of } t: (t + 4) \cdot \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0.$$

Assume a power series solution centered at $x_0 = 4$: $y(t) = a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots$

First we need to find $y'(x)$ and $y''(x)$.

Hence, $y'(t) = a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots$ and $y''(t) = 2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots$

Substituting $y(t)$, $y'(t)$, and $y''(t)$ into the differential equation $(t+4) \cdot \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0$:

$$(t+4)(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + (a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0$$

$$\Rightarrow t(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + 4(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + (a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0$$

$$\Rightarrow (2a_2t + 2 \cdot 3a_3t^2 + 3 \cdot 4a_4t^3 + 4 \cdot 5a_5t^4 + \dots) + (4 \cdot 2a_2 + 4 \cdot 2 \cdot 3a_3t + 4 \cdot 3 \cdot 4a_4t^2 + 4 \cdot 4 \cdot 5a_5t^3 + \dots) + (a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0 + 0t + 0t^2 + 0t^3 + 0t^4 + 0t^5 + \dots$$

Comparing the coefficients of corresponding powers of t on both sides:

$$\text{For } t^0 : 4 \cdot 2a_2 + a_1 + a_0 = 0$$

$$\text{For } t^1 : 2a_2 + 4 \cdot 2 \cdot 3a_3 + 2a_2 + a_1 = 0$$

$$\text{For } t^2 : 2 \cdot 3a_3 + 4 \cdot 3 \cdot 4a_4 + 3a_3 + a_2 = 0$$

$$\text{For } t^3 : 3 \cdot 4a_4 + 4 \cdot 4 \cdot 5a_5 + 4a_4 + a_3 = 0$$

$$\text{For } t^4 : 4 \cdot 5a_5 + 4 \cdot 5 \cdot 6a_6 + 5a_5 + a_4 = 0$$

$$\text{For } t^n : n \cdot (n+1)a_{n+1} + 4 \cdot (n+1) \cdot (n+2) \cdot a_{n+2} + (n+1)a_{n+1} + a_n = 0$$

$$\Rightarrow 4 \cdot (n+1) \cdot (n+2) \cdot a_{n+2} = -[(n+1)a_{n+1} + a_n + n \cdot (n+1)a_{n+1}]$$

$$\Rightarrow a_{n+2} = \frac{-[(n+1+n \cdot (n+1))a_{n+1} + a_n]}{4 \cdot (n+1) \cdot (n+2)}$$

$$\text{Recurrence relation is } a_{n+2} = -\frac{[(n+1+n \cdot (n+1))a_{n+1} + a_n]}{4 \cdot (n+1) \cdot (n+2)} = -\frac{[(n^2 + 2n + 1)a_{n+1} + a_n]}{4 \cdot (n+1) \cdot (n+2)}$$

$$\text{For } n = 0 : a_2 = -\frac{[(0^2 + 2 \cdot 0 + 1)a_1 + a_0]}{4 \cdot (0+1) \cdot (0+2)} = -\frac{a_0 + a_1}{8} = -\frac{1}{8}a_0 - \frac{1}{8}a_1$$

$$\text{For } n=1: a_3 = -\frac{\left[(1^2 + 2 \cdot 1 + 1)a_2 + a_1\right]}{4 \cdot (1+1) \cdot (1+2)} = -\frac{4a_2 + a_1}{24} = -\frac{4\left[-\frac{1}{8}a_0 - \frac{1}{8}a_1\right] + a_1}{24} = \frac{1}{48}a_0 - \frac{1}{48}a_1$$

$$\text{For } n=2: a_4 = -\frac{\left[(2^2 + 2 \cdot 2 + 1)a_3 + a_2\right]}{4 \cdot (2+1) \cdot (2+2)} = -\frac{9a_3 + a_2}{48} = -\frac{9\left[\frac{1}{48}a_0 - \frac{1}{48}a_1\right] + a_2}{48} = -\frac{1}{768}a_0 + \frac{5}{768}a_1$$

a) The singular point of the differential equation $x \cdot y'' + y' + y = 0$ is ____?

b) The power series solution for $x \cdot y'' + y' + y = 0$, centered at $x_0 = 4$, is $y(x) = a_0 \cdot [\text{____?}] + a_1 \cdot [\text{____?}]$

Hint: The power series solution centered at $x_0 = 4$ is $y(t) = a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots$

$$y(x) = a_0 + a_1(x-4)x + a_2(x-4)x^2 + a_3(x-4)x^3 + a_4(x-4)x^4 + a_5(x-4)x^5 + \dots$$

c) Radius of convergence for power series solution is ____?

d) Interval of convergence for power series solution is ____?

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

Question 6:

(This problem is optional.)

Find the power series solution for the differential equation $(x^2 - 9)y'' + y' + 4y = 0$, centered at the ordinary point $x_0 = 2$:

Since we are looking for a power series solution centered at $x_0 = 2$, define $t = x - 2$.

Hence, $x = t + 2$ and $x^2 - 9 = (t + 2)^2 - 9 = t^2 + 4t + 4 - 9 = t^2 + 4t - 5$.

$$\text{Note: } \frac{dx}{dt} = 1, \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{dy}{dt}, \text{ and } \frac{d^2y}{dx^2} = \frac{d}{dx} \cdot \left[\frac{dy}{dx} \right] = \frac{d \left[\frac{dy}{dx} \right] / dt}{dx/dt} = \frac{d \left[\frac{dy}{dt} \right] / dt}{1} = \frac{d^2y}{dt^2}.$$

Write $(x^2 - 9)y'' + y' + 4y = 0$ in terms of t : $(t^2 + 4t - 5) \cdot \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0$.

Assume a power series solution centered at $x_0 = 2$ is $y(t) = a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots$

First we need to find $y'(x)$ and $y''(x)$.

Hence, $y'(t) = a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots$ and $y''(t) = 2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots$

Substituting $y(t)$, $y'(t)$, and $y''(t)$ into the differential equation $(t^2 + 4t - 5) \cdot \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0$:

$$(t^2 + 4t - 5)(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + (a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0$$

$$\Rightarrow t^2(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + 4t(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) - 5(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + (a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0$$

$$\Rightarrow (2a_2t^2 + 2 \cdot 3a_3t^3 + 3 \cdot 4a_4t^4 + 4 \cdot 5a_5t^5 + \dots) + (4 \cdot 2a_2 \cdot t + 4 \cdot 2 \cdot 3a_3t^2 + 4 \cdot 3 \cdot 4a_4t^3 + 4 \cdot 4 \cdot 5a_5t^4 + \dots) - (5 \cdot 2a_2 + 5 \cdot 2 \cdot 3a_3t + 5 \cdot 3 \cdot 4a_4t^2 + 5 \cdot 4 \cdot 5a_5t^3 + \dots) + (a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0$$

$$\Rightarrow (1 \cdot 2a_2t^2 + 2 \cdot 3a_3t^3 + 3 \cdot 4a_4t^4 + 4 \cdot 5a_5t^5 + \dots) + (4 \cdot 2a_2 \cdot t + 4 \cdot 2 \cdot 3a_3t^2 + 4 \cdot 3 \cdot 4a_4t^3 + 4 \cdot 4 \cdot 5a_5t^4 + \dots) - (5 \cdot 2a_2 \cdot t^0 + 5 \cdot 2 \cdot 3a_3t + 5 \cdot 3 \cdot 4a_4t^2 + 5 \cdot 4 \cdot 5a_5t^3 + \dots) + (1 \cdot a_1 \cdot t^0 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 \cdot t^0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0 + 0t + 0t^2 + 0t^3 + 0t^4 + 0t^5 + \dots$$

Comparing the coefficients of corresponding powers of t on both sides:

$$\text{For } t^0: -5 \cdot 2a_0 + 1 \cdot a_1 + a_0 = 0$$

$$\text{For } t^1: 4 \cdot 1 \cdot 2a_2 - 5 \cdot 2 \cdot 3a_3 + 2a_2 + a_1 = 0$$

$$\text{For } t^2: 1 \cdot 2a_2 + 4 \cdot 2 \cdot 3a_3 - 5 \cdot 3 \cdot 4a_4 + 3a_3 + a_2 = 0$$

$$\text{For } t^3: 2 \cdot 3a_3 + 4 \cdot 3 \cdot 4a_4 - 5 \cdot 4 \cdot 5a_5 + 4a_4 + a_3 = 0$$

$$\text{For } t^4: 3 \cdot 4a_4 + 4 \cdot 4 \cdot 5a_5 - 5 \cdot 5 \cdot 6a_6 + 5a_5 + a_4 = 0$$

$$\text{For } t^n: (n-1) \cdot n \cdot a_n + 4 \cdot n \cdot (n+1) \cdot a_{n+1} - 5 \cdot (n+1) \cdot (n+2) a_{n+2} + (n+1) a_{n+1} + a_n = 0$$

$$\text{For } t^n: (n-1) \cdot n \cdot a_n + 4 \cdot n \cdot (n+1) \cdot a_{n+1} + (n+1) a_{n+1} + a_n = 5 \cdot (n+1) \cdot (n+2) a_{n+2}$$

$$\text{Recurrence relation is } a_{n+2} = \frac{(n-1) \cdot n \cdot a_n + 4 \cdot n \cdot (n+1) \cdot a_{n+1} + (n+1) a_{n+1} + a_n}{5 \cdot (n+1) \cdot (n+2)}$$

$$\text{For } n=0: a_2 = \frac{(0-1) \cdot (0) \cdot a_0 + 4 \cdot (0) \cdot (0+1) \cdot a_1 + (0+1) a_1 + a_0}{5 \cdot (0+1) \cdot (0+2)} = \frac{a_1 + a_0}{10} = \frac{1}{10} a_0 + \frac{1}{10} a_1$$

$$\text{For } n=1: a_3 = \frac{[(1-1) \cdot 1 + 1] \cdot a_1 + [4 \cdot 1 \cdot (1+1) + (1+1)] a_2}{5 \cdot (1+1) \cdot (1+2)} = \frac{a_1 + 10a_2}{30} = \frac{a_1 + 10 \left[\frac{1}{10} a_0 + \frac{1}{10} a_1 \right]}{30} = \frac{a_0 + 2a_1}{30} = \frac{1}{30} a_0 + \frac{1}{15} a_1$$

$$\text{For } n = 2: a_4 = \frac{[(2-1) \cdot 2 + 1] \cdot a_2 + [4 \cdot 2 \cdot (2+1) + (2+1)] a_3}{5 \cdot (2+1) \cdot (2+2)} = \frac{3a_2 + 27a_3}{60} = \frac{3 \left[\frac{1}{10} a_0 + \frac{1}{10} a_1 \right] + 27 \left[\frac{1}{30} a_0 + \frac{1}{15} a_1 \right]}{60} = \frac{1}{50} a_0 + \frac{7}{2000} a_1$$

a) The singular points of the differential equation $(x^2 - 9)y'' + y' + 4y = 0$ are ____?

b) The power series solution for $(x^2 - 9)y'' + y' + 4y = 0$, centered at $x_0 = 2$ is $y(x) = a_0 \cdot [\text{____?}] + a_1 \cdot [\text{____?}]$

Hint: The power series solution centered at $x_0 = 2$ is $y(t) = a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots$

$$y(x) = a_0 + a_1(x-4)x + a_2(x-4)x^2 + a_3(x-4)x^3 + a_4(x-4)x^4 + a_5(x-4)x^5 + \dots$$

c) Radius of convergence for power series solution is ____?

d) Interval of convergence for power series solution is ____?

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.

Question 7: (This problem is optional.)

Find the power series solution for the differential equation $x^2 \cdot y'' + x \cdot y' + y = 0$, centered at the ordinary point $x_0 = 3$.

Since we are looking for a power series solution centered at $x_0 = 3$, define $t = x - 3$.

Hence, $x = t + 3$ and $x^2 = (t + 3)^2 = t^2 + 6t + 9$.

Note: $\frac{dx}{dt} = 1$, $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{dy}{dt}$, and $\frac{d^2y}{dx^2} = \frac{d^2y}{dt^2}$.

Write $x^2 \cdot y'' + x \cdot y' + y = 0$ in terms of t : $(t^2 + 6t + 9) \cdot \frac{d^2y}{dt^2} + (t + 3) \cdot \frac{dy}{dt} + y = 0$.

Assume a power series solution centered at $t = 3$: $y(t) = a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots$

Hence, $y'(t) = a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots$ and $y''(t) = 2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots$

Substituting $y(t)$, $y'(t)$, and $y''(t)$ into the differential equation $(t^2 + 6t + 9) \cdot \frac{d^2y}{dt^2} + (t + 3) \cdot \frac{dy}{dt} + y = 0$:

$$(t^2 + 6t + 9)(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + (t + 3)(a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0$$

$$\Rightarrow t^2(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + 6t(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) + 9(2a_2 + 2 \cdot 3a_3t + 3 \cdot 4a_4t^2 + 4 \cdot 5a_5t^3 + \dots) \\ + t(a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + 3(a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4 + \dots) + (a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0$$

$$\Rightarrow (2a_2t^2 + 2 \cdot 3a_3t^3 + 3 \cdot 4a_4t^4 + 4 \cdot 5a_5t^5 + \dots) + (6 \cdot 2a_2 \cdot t + 6 \cdot 2 \cdot 3a_3t^2 + 6 \cdot 3 \cdot 4a_4t^3 + 6 \cdot 4 \cdot 5a_5t^4 + \dots) + (9 \cdot 2a_2 + 9 \cdot 2 \cdot 3a_3t + 9 \cdot 3 \cdot 4a_4t^2 + 9 \cdot 4 \cdot 5a_5t^3 + \dots) \\ + (a_1 \cdot t^1 + 2a_2t^2 + 3a_3t^3 + 4a_4t^4 + 5a_5t^5 + \dots) + (3a_1 + 3 \cdot 2a_2t + 3 \cdot 3a_3t^2 + 3 \cdot 4a_4t^3 + 3 \cdot 5a_5t^4 + \dots) + (a_0 \cdot t^0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots) = 0$$

Comparing the coefficients of corresponding powers of t on both sides:

$$\text{For } t^0: 9 \cdot 2a_2 + 2a_1 + 1 \cdot a_1 + a_0 = 0$$

$$\text{For } t^1: 6 \cdot 2a_2 + 9 \cdot 2 \cdot 3a_3 + a_1 + 3 \cdot 2a_2 + a_1 = 0$$

$$\text{For } t^2: 2a_2 + 6 \cdot 2 \cdot 3a_4 + 9 \cdot 3 \cdot 4a_4 + 2a_2 + 3 \cdot 3a_3 + a_2 = 0$$

$$\text{For } t^3: 2 \cdot 3a_3 + 6 \cdot 3 \cdot 4a_4 + 9 \cdot 4 \cdot 5a_5 + 3a_3 + 3 \cdot 4a_4 + a_3 = 0$$

$$\text{For } t^4: 3 \cdot 4a_4 + 6 \cdot 4 \cdot 5a_5 + 9 \cdot 5 \cdot 6a_6 + 4a_4 + 3 \cdot 5a_5 + a_4 = 0$$

$$\text{For } t^n: (n-1) \cdot n \cdot a_n + 6 \cdot n \cdot (n+1) \cdot a_{n+1} + 9 \cdot (n+1) \cdot (n+2) a_{n+2} + na_n + 3 \cdot (n+1) a_{n+1} + a_n = 0$$

$$\Rightarrow 9 \cdot (n+1) \cdot (n+2) a_{n+2} = -[(n-1) \cdot n \cdot a_n + 6 \cdot n \cdot (n+1) \cdot a_{n+1} + na_n + 3 \cdot (n+1) a_{n+1} + a_n]$$

$$\Rightarrow a_{n+2} = \frac{-[(n-1) \cdot n \cdot a_n + 6 \cdot n \cdot (n+1) \cdot a_{n+1} + na_n + 3 \cdot (n+1) a_{n+1} + a_n]}{9 \cdot (n+1) \cdot (n+2)} = -\frac{[(n-1) \cdot n + n + 1] a_n + [6 \cdot n \cdot (n+1) + 3 \cdot (n+1)] \cdot a_{n+1}}{9 \cdot (n+1) \cdot (n+2)}$$

$$\text{Recurrence relation is } a_{n+2} = -\frac{[(n-1) \cdot n + n + 1] a_n + [6 \cdot n \cdot (n+1) + 3 \cdot (n+1)] \cdot a_{n+1}}{9 \cdot (n+1) \cdot (n+2)}$$

$$\text{For } n=0: a_2 = -\frac{[(0-1) \cdot 0 + 0 + 1] a_0 + [4 \cdot 0 \cdot (0+1) + 3 \cdot (0+1)] \cdot a_1}{9 \cdot (0+1) \cdot (0+2)} = -\frac{a_0 + 3a_1}{18}$$

$$\text{For } n=1: a_3 = -\frac{[(1-1) \cdot 1 + 1 + 1]a_1 + [6 \cdot 1 \cdot (1+1) + 3 \cdot (1+1)] \cdot a_2}{9 \cdot (1+1) \cdot (1+2)} = -\frac{2a_1 + 18a_2}{54} = -\frac{2a_1 + 18\left[-\frac{a_0 + 3a_1}{18}\right]}{54} = \frac{a_0 + a_2}{54} = \frac{a_0}{54} + \frac{a_1}{54}$$

$$\text{For } n=2: a_4 = -\frac{[(2-1) \cdot 2 + 2 + 1]a_2 + [6 \cdot 2 \cdot (2+1) + 3 \cdot (2+1)] \cdot a_3}{9 \cdot (2+1) \cdot (2+2)} = -\frac{5a_2 + 45a_3}{108} = -\frac{5\left[-\frac{a_0 + 3a_1}{18}\right] + 45\left[\frac{a_0}{54} + \frac{a_1}{54}\right]}{108} = -\frac{5}{972}a_0 + 0a_1$$

$$\text{For } n=3: a_5 = -\frac{[(3-1) \cdot 3 + 3 + 1]a_3 + [6 \cdot 3 \cdot (3+1) + 3 \cdot (3+1)] \cdot a_4}{9 \cdot (3+1) \cdot (3+2)} = -\frac{10a_2 + 84a_3}{180} = \frac{1}{792}a_0 - \frac{1}{792}a_1$$

a) The singular points of the differential equation $x^2y'' + xy' + y = 0$ are ____?

b) The power series solution for $x^2y'' + xy' + y = 0$, centered at $x_0 = 3$ is $y(x) = a_0 \cdot [\text{____?}] + a_1 \cdot [\text{____?}]$

Hint: The power series solution centered at $x_0 = 3$ is $y(t) = a_0 + a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + a_4 \cdot t^4 + \dots$

$$y(x) = a_0 + a_1(x-4)x + a_2(x-4)x^2 + a_3(x-4)x^3 + a_4(x-4)x^4 + a_5(x-4)x^5 + \dots$$

c) Radius of convergence for power series solution is ____?

d) Interval of convergence for power series solution is ____?

Note: Fuchs's theorem (related to Frobenius method)

The radius of convergence is the minimum Euclidian distance

between x_0 and the nearest singular point;

x_0 is the point where the power series solution is centered.

Interval of convergence is $(x_0 - R, x_0 + R)$; R is radius of convergence.